HEILLON

COMPUTATIONAL LEGITIMACY INFRASTRUCTURE

Technical Whitepaper

v2.0

STATUS · CONSTITUTIONAL_ACTIVE

VERSION · v3.5.0 · Schema M2+M3+Guard+PULSE+HSC

CCR-03 · 27/27 PASS · a3f9e2c1b8d46f50...

CORPUS · 13 norms · BR / UAE / EU

DATE · March 2026 · Joinville, SC, Brazil

INVISIBLE AT INSTALLATION. INEVITABLE AS PROOF.

01

Executive Summary

HEILLON is the world's first **Computational Legitimacy Infrastructure** — a Decision Verification Layer that makes every automated and AI-driven decision auditable, sovereign, and irrefutable. It operates between autonomous systems and institutional infrastructure as the only layer that can structurally guarantee legitimacy at machine speed.

CORE VALUE PROPOSITION

HEILLON makes legitimacy a *computable, verifiable property* of every execution — not a post-hoc claim, not a log, not a compliance checkbox. A structural proof, sealed before the decision executes.

HEILLON is not a logging system. Not an audit tool. Not a governance dashboard. It is **infrastructure** — the missing constitutional layer between AI systems and the institutions, regulators, and courts that must hold them accountable.

3	13	<60s
VERIFIABLE DIMENSIONS per decision	NORMS AUTO-BOUND BR + UAE + EU	FROM ZERO TO FIRST HDR

02

The Governance Gap

AI systems now trigger workflows, execute financial transactions, and make operational decisions at machine speed. Institutional governance was designed for humans. The structural gap between the two is not a process problem — it is an infrastructure problem.

PROBLEM 01

Logs are not proof

Logs record what happened. They do not prove who had authority to act, under what mandate, or whether policy boundaries were respected at the moment of execution. Regulators require proof.

PROBLEM 02

Accountability collapses at speed

When an AI approves a $40M credit line at 3AM, initiates treatment for 12,000 patients, or recommends policy affecting 2M citizens — no existing layer can answer: was this legitimate?

PROBLEM 03

Regulatory exposure is now

EU AI Act Art.12 obligations for high-risk systems take effect August 2026. Fines up to €30M or 6% of global turnover. BCB 4.557 and LGPD Art.20 are already in force in Brazil.

> **HEILLON resolves this structurally.** Every decision enters the system as a declared intention, passes through the Z-J-M-H-K authority pipeline, and exits as a cryptographically sealed proof — permanently reconstructable by any auditor, regulator, or court at any future point.

HDR v2.0 — The Heillon Decision Record

The HDR is the primary auditable artifact. Version 2.0 introduces three independent verifiable dimensions in a single cryptographic seal — the only infrastructure in the world that proves all three simultaneously.

D1 · PROOF OF FACT

What happened, when, with which hash

Cryptographic proof that a specific event occurred at a verifiable timestamp. Tamper any field and the SHA-256 chain breaks. The fact is irrefutable.

→ decision_id
→ execution_hash (SHA-256)
→ timestamp_sovereign (HSC)
→ chain_position

D2 · PROOF OF AUTHORITY

Who was authorized at the exact moment

Cryptographic verification of the authority chain at execution time. Proves mandate scope, NLO level, and Z-J-M-H-K pipeline compliance.

→ authority_id
→ mandate_hash
→ nlo_level (1-5)
→ pipeline: Z-J-M-H-K

D3 · NORMATIVE CONTEXT

Which norms were in force at decision time

D3 does not ask what the regulation says today. It proves what was in force at the moment of the decision — 13 norms automatically bound at timestamp_sovereign.

→ norms[] (auto-bound)
→ jurisdiction
→ corpus_id
→ binding_at_timestamp

Open protocol · RFC-0001 CANONICAL · RFC-0002 PUBLISHED · RFC-0003 DRAFT · Verify at hdr.heillon.com

04

Architecture: Five Pillars + Z-J-M-H-K Pipeline

THE FIVE PILLARS

P-01 **Sovereign Time**
RFC 3161 tamper-proof timestamps. No retroactive manipulation.

P-02 **Immutable Ledger**
SHA-256 hash chain. Tamper any event, chain breaks permanently.

P-03 **Formal Authority**
Actions bound to verifiable mandate chains. Always traceable.

P-04 **Prior Intention**
Intent

```
<div>
  <div style="font-family: 'Courier New', monospace; font-size: 9px; letter-spacing: 2px; color: #7b7fc4; margin-bottom: 12px;">THE Z-J-M-H-K PIPELINE</div>
  <div style="display: flex; flex-direction: column; gap: 1px; background: #08080A;">
    <div style="background: #0c0e18; padding: 10px 14px; display: grid; grid-template-columns: 22px 1fr; gap: 8px;"><span style="font-family:'Courier New',monospace; font-size:14px; font-weight:bold; color:#C9A84C; line-height:1;">Z</span><div><strong style="font-size:11px; color:#F0EBE0;">Zone</strong><br><span style="font-size:10px; color:#7b7fc4;">Institutional domain + boundaries defined</span></div></div>
    <div style="background: #0c0e18; padding: 10px 14px; display: grid; grid-template-columns: 22px 1fr; gap:
```

declared and sealed BEFORE execution. Never inferred.

P-05 **NLO Levels 1–5**

Certifiable legitimacy scores. Sold as contractual products.

```
8px;"><span style="font-
family:'Courier New',monospace; font-
size:14px; font-weight:bold;
color:#C9A84C; line-
height:1;">J</span><div><strong
style="font-size:11px;
color:#F0EBE0;">Jurisdiction</strong>
<br><span style="font-size:10px;
color:#7b7fc4;">Cryptographic
authority verified</span></div></div>
    <div style="background: #0c0e18;
padding: 10px 14px; display: grid;
grid-template-columns: 22px 1fr; gap:
8px;"><span style="font-
family:'Courier New',monospace; font-
size:14px; font-weight:bold;
color:#C9A84C; line-
height:1;">M</span><div><strong
style="font-size:11px;
color:#F0EBE0;">Mandate</strong><br>
<span style="font-size:10px;
color:#7b7fc4;">Prior intention
sealed to ledger before action</span>
</div></div>
    <div style="background: #C9A84C;
padding: 10px 14px; display: grid;
grid-template-columns: 22px 1fr; gap:
8px;"><span style="font-
family:'Courier New',monospace; font-
size:14px; font-weight:bold;
color:#08080A; line-
height:1;">H</span><div><strong
style="font-size:11px;
color:#08080A;">HEILLON SEAL</strong>
```

```
<br><span style="font-size:10px;
color:#3d2a00;">Execution +
constitutional hash + HDR
generated</span></div></div>
    <div style="background: #0c0e18;
padding: 10px 14px; display: grid;
grid-template-columns: 22px 1fr; gap:
8px;"><span style="font-
family:'Courier New',monospace; font-
size:14px; font-weight:bold;
color:#C9A84C; line-
height:1;">K</span><div><strong
style="font-size:11px;
color:#F0EBE0;">Knowledge</strong>
<br><span style="font-size:10px;
color:#7b7fc4;">NLO certification +
audit record generated</span></div>
</div>
  </div>
</div>
```

No decision reaches execution without passing all five pipeline stages. Intent is NEVER inferred — only DECLARED or OBSERVED.

05

HPG, HAV & Trust Engine

```
<div style="background:
#08080A; padding: 20px;">
  <div style="font-
family:'Courier
New',monospace; font-
size:9px; letter-
spacing:2px;
color:#7b7fc4; margin-
bottom:8px;">HPG · HEILLON
PULSE GRAVITY</div>
  <div style="font-
size:14px; font-
weight:300; color:#F0EBE0;
margin-bottom:10px;">The
invisible layer.
Inevitable as gravity.
</div>
  <div style="font-
size:11px; color:#a0a8b3;
line-height:1.7; margin-
bottom:12px;">HPG operates
inside your infrastructure
without modifying it. One
endpoint. Every decision
becomes a
cryptographically verified
institutional artifact.
Zero instrumentation of
```

```
the existing system.</div>
  <div style="font-
family:'Courier
New',monospace; font-
size:9px; color:#40916C;
line-height:1.9;">
    POST /pulse/onboard →
client_id<br>
    POST /pulse/declare →
signal_id + confidence<br>
    GET /certificate →
cryptographic PDF
  </div>
</div>

<div style="display:flex;
flex-direction:column;
gap:12px;">
  <div
style="background:#f0faf4;
padding:16px; border-
left:3px solid #2D6A4F;">
    <div style="font-
family:'Courier
New',monospace; font-
size:9px; letter-
spacing:2px;
color:#2D6A4F; margin-
bottom:6px;">HAV · HEILLON
AUDIT VIGIL</div>
    <div style="font-
size:12px; font-
weight:bold;
color:#1a1a1a; margin-
```

```
bottom:6px;">Sovereign
programmable audit
scheduler</div>
    <div style="font-
size:11px; color:#555;
line-height:1.6;">The
client defines audit type,
frequency, scope, and
delivery format. HAV
executes autonomously,
generates a sealed HDR +
report + integrity hash —
without human
intervention.</div>
  </div>
  <div
style="background:#f0f2ff;
padding:16px; border-
left:3px solid #3d3d8f;">
    <div style="font-
family:'Courier
New',monospace; font-
size:9px; letter-
spacing:2px;
color:#3d3d8f; margin-
bottom:6px;">HSC · HEILLON
SOVEREIGN CHRONOS</div>
    <div style="font-
size:12px; font-
weight:bold;
color:#1a1a1a; margin-
bottom:6px;">Sovereign
time — pillar H of Z-J-M-
H-K</div>
```

```
    <div style="font-size:11px; color:#555; line-height:1.6;">Every decision carries an irrefutable timestamp from a sovereign source. Mandatory at launch. Resolves temporal conflict in distributed systems. RFC 3161-compatible.</div>
  </div>
</div>
```

TRUST ENGINE · P(D|S) — CONFIDENCE IS CALCULATED, NOT ASSUMED

$$P(D|S) \approx 1 - \prod(1 - w_i \cdot P(s_i))$$

SOVEREIGN	VERIFIED	ANCHORED	INFERRED	UNTRUSTED
≥ 0.99	≥ 0.85	≥ 0.70	≥ 0.50	< 0.50

Absolute rule: intent is NEVER inferred — only DECLARED or OBSERVED. ML may only enrich context, never determine intent.

06

Normative Corpus — 13 Norms, 3 Jurisdictions

D3 (Normative Context) does not ask what the regulation says today. It proves what was in force at the *timestamp_sovereign* of the decision. The corpus binds automatically — no manual configuration required.

```
<div
style="background:white;
padding:20px;">
  <div style="font-
size:20px; margin-
bottom:6px;">BR</div>
  <div style="font-
family:'Courier
New',monospace; font-
size:9px; letter-
spacing:2px;
color:#8B4513; margin-
bottom:12px;">BRAZIL · 7
NORMS</div>
  <div style="font-
size:11px; color:#333;
line-height:2.0;">
    <strong>BCB Res.
4.557</strong> — AI risk
governance<br>
    <strong>LGPD Art.
7</strong> — Data
processing grounds<br>
```

```
    <strong>LGPD Art. 18</strong> — Data subject rights<br>
    <strong>LGPD Art. 20</strong> — Automated decisions<br>
    <strong>CMN Res. 4.893</strong> — Cyber risk<br>
    <strong>CVM Res. 80</strong> — AI in capital markets<br>
    <strong>Marco Civil Art. 7</strong> — Digital rights
  </div>
</div>

<div style="background:white; padding:20px;">
  <div style="font-size:20px; margin-bottom:6px;">AE</div>
  <div style="font-family:'Courier New',monospace; font-size:9px; letter-spacing:2px; color:#1e5799; margin-bottom:12px;">UAE / DIFC · 3 NORMS</div>
  <div style="font-size:11px; color:#333;
```

```
line-height:2.0;">
    <strong>DIFC Reg.
10</strong> — AI
decision
traceability<br>
    <strong>UAE AI
Policy 2031</strong> —
AI governance
framework<br>
    <strong>DFSA
COB</strong> — Conduct
of business rules
  </div>
  <div style="margin-
top:16px; font-
size:10px; color:#555;
line-height:1.6;">DIFC
Reg.10 is structurally
aligned with HDR v2.0.
UAE is the highest-
priority international
market — AI in Cabinet
Jan/2026, zero local
player in this space.
</div>
</div>

<div
style="background:white;
padding:20px;">
  <div style="font-
size:20px; margin-
bottom:6px;">EU</div>
  <div style="font-
```

```
family:'Courier
New',monospace; font-
size:9px; letter-
spacing:2px;
color:#6b3fa0; margin-
bottom:12px;">EUROPEAN
UNION · 3 NORMS</div>
  <div style="font-
size:11px; color:#333;
line-height:2.0;">
    <strong>EU AI Act
Art. 9</strong> — Risk
management systems<br>
    <strong>EU AI Act
Art. 12</strong> —
Logging & audit
trails<br>
    <strong>GDPR Art.
5/22</strong> —
Automated decision
rights
  </div>
  <div style="margin-
top:16px; font-
size:10px; color:#555;
line-height:1.6;">Art.12
obligations for high-
risk AI systems take
effect August 2026.
HEILLON provides the
cryptographic audit
trail that compliance
requires — beyond simple
```

```
logging.</div>
</div>
```

07

Current System State & Integration

```
<div>
  <div style="font-
family:'Courier New',monospace;
font-size:9px; letter-
spacing:2px; color:#7b7fc4;
margin-bottom:10px;">PRODUCTION
STATUS · MARCH 2026</div>
  <div
style="background:#08080A;
padding:16px; font-
family:'Courier New',monospace;
font-size:10px; line-
height:2.0;">
    <div
style="color:#40916C;">●
CONSTITUTIONAL_ACTIVE</div>
    <div style="color:#C9A84C;">
Version        v3.5.0</div>
    <div style="color:#C9A84C;">
Schema
M2+M3+Guard+PULSE+HSC</div>
    <div style="color:#C9A84C;">
Tables         18 active</div>
    <div style="color:#C9A84C;">
CCR-03         27/27 PASS</div>
    <div style="color:#C9A84C;">
Hash
a3f9e2c1b8d46f50...</div>
```

```
    <div style="color:#C9A84C;">
Backend        Railway ·
persistent volume</div>
    <div style="color:#C9A84C;">
Portal         heillon.com ·
roles active</div>
    <div style="color:#C9A84C;">
Corpus         13 norms
bound</div>
    <div style="color:#40916C;">
Verifier       public ·
hdr.heillon.com</div>
  </div>
</div>

<div>
  <div style="font-
family:'Courier New',monospace;
font-size:9px; letter-
spacing:2px; color:#7b7fc4;
margin-bottom:10px;">ZERO-
FRICTION INTEGRATION</div>
  <div style="font-size:12px;
color:#333; margin-bottom:12px;
line-height:1.7;">HEILLON
integrates in 3 API calls. Your
system does not change. No
refactoring. No SDK
installation. No production data
required for POC.</div>
  <div
style="background:#0c0e18;
padding:14px; font-
family:'Courier New',monospace;
```

```
font-size:9px; color:#40916C;
line-height:2.0;">
    # Step 1: Onboard<br>
    POST /pulse/onboard<br>
    <span
style="color:#7b7fc4;">→
client_id + integration
URLs</span><br><br>
    # Step 2: Declare each
decision<br>
    POST /pulse/declare<br>
    <span
style="color:#7b7fc4;">{
actor_id, action, subject_id }
</span><br>
    <span
style="color:#7b7fc4;">→
signal_id + confidence
level</span><br><br>
    # Step 3: Retrieve proof<br>
    GET
/signal/{'{id}'}/certificate<br>
    <span
style="color:#7b7fc4;">→
cryptographic PDF · offline-
verifiable</span>
  </div>
  <div style="font-size:11px;
color:#2D6A4F; font-weight:bold;
margin-top:8px;">✓ First HDR in
under 60 seconds from zero.
</div>
</div>
```

ACCESS & PORTAL

The HEILLON portal (**heillon.com**) operates with sovereign access codes (HEILLON-XXXX-XXXX) and role-based routing. Roles determine destination automatically: HEILLON_COMMERCIAL and CLIENT_ANALYST access the institutional portal; HEILLON_OPERATOR and CLIENT_OPERATOR are routed directly to the Command Center. No passwords. No forms.

08

Market Position & Seed Round

GO-TO-MARKET SEQUENCE

PHASE 1 · NOW → 6M

Brazil Fintechs — LGPD Art.20 + BCB 4.557 + CVM 80. Seed round closing. First 3–5 POC clients. MOU targeting.

PHASE 2 · 6–18M

UAE / DIFC — DIFC Reg.10 aligned. AI in Cabinet Jan/2026. Stargate 5GW. Zero local player. Disruption level 1 certain.

PHASE 3 · 12–24M

EU — EU AI Act Art.12 August 2026 deadline. GDPR Art.22. Institutional and enterprise market.

```
<div>
  <div style="font-family:'Courier New',monospace; font-size:9px; letter-spacing:2px; color:#7b7fc4; margin-bottom:12px;">SEED ROUND</div>
  <div style="display:grid; grid-template-columns:1fr 1fr; gap:8px; margin-bottom:16px;">
    <div style="background:#f8f6f0; padding:14px; border-top:2px solid #C9A84C; text-align:center;"><div style="font-family:'Courier New',monospace; font-size:16px; color:#C9A84C; font-weight:bold;">$1.5–3M</div><div style="font-family:'Courier New',monospace; font-size:8px; color:#666;
```

```
margin-top:4px;">RAISE
TARGET</div></div>
    <div
style="background:#f8f6f0;
padding:14px; border-
top:2px solid #C9A84C;
text-align:center;"><div
style="font-
family:'Courier
New',monospace; font-
size:16px; color:#C9A84C;
font-weight:bold;">$5–
7M</div><div style="font-
family:'Courier
New',monospace; font-
size:8px; color:#666;
margin-top:4px;">PRE-
MONEY</div></div>
    <div
style="background:#f8f6f0;
padding:14px; border-
top:2px solid #2D6A4F;
text-align:center;"><div
style="font-
family:'Courier
New',monospace; font-
size:16px; color:#2D6A4F;
font-
weight:bold;">$668B</div>
<div style="font-
family:'Courier
New',monospace; font-
size:8px; color:#666;
margin-top:4px;">TAM AI
```

```
GOVERNANCE</div></div>
    <div
style="background:#f8f6f0;
padding:14px; border-
top:2px solid #2D6A4F;
text-align:center;"><div
style="font-
family:'Courier
New',monospace; font-
size:16px; color:#2D6A4F;
font-
weight:bold;">18M</div>
<div style="font-
family:'Courier
New',monospace; font-
size:8px; color:#666;
margin-top:4px;">TO EU AI
ACT DEADLINE</div></div>
  </div>
  <div style="font-
size:11px; color:#555;
line-height:1.7;
background:#f0faf4;
padding:12px; border-
left:2px solid #2D6A4F;">
    <strong>Why now:
</strong> EU AI Act
deadline August 2026
creates a hard regulatory
clock. BCB and CVM
enforcement is already
active in Brazil. UAE is
accelerating AI governance
with no infrastructure
```

```
player in place. HEILLON
is the only infrastructure
that covers all three
jurisdictions
simultaneously with live
production proof.
  </div>
</div>
```

OPEN PROTOCOL

hdr.heillon.com

PRODUCT DEMO

vitrine.heillon.com

CONTACT

contact@heillon.com

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